Crop Production

Kharif Crop – Grown in **summer season**

Example – Paddy, Maize, Jowar, Bajra, groundnut, til (Sesame), Cotton, sun hemp, moong, urd, soyabean

Rabi crop – Grown in **winter season**

Example – Wheat, Barley, Gram, Mustard, Sugarcane, Tobacco, Berseem.

Kharif/Rainy/Monsoon crops	Rabi/winter/cold seasons crops	Summer/Zaid crops
The crops grown in **monsoon** months	The crops grown in **winter** season	Crops grown in **summer**
Sown **before monsoon and harvested at the end of the monsoon**	**Sown before retreating monsoon** and harvested before summer.	**Sown and harvested in summer**
June to Oct-Nov	Oct to March	March to June
Require **warm, wet weather** at major period of crop growth	Crops grow **well in cold** and dry weather	Require **warm dry weather** for major growth period
E.g. Cotton, Rice, Jowar, Bajara etc.	**E.g. Wheat, gram, sunflower etc.**	**E.g. Groundnuts, Watermelon, Pumpkins, Gourds etc.**

Classification of crops

- **Cereal crop** – Rice, wheat, Maize, Sorghum, bajra (Pearl millet), Barley, Ragi, Kodo
- **Pulse crop** – Gram, Lentil, Pea, Arhar, Pea, Blackgram, Cowpea, Lobia
- **Oilseed crop** – Groundnut, seasmum, castor, Rapeseed, linseed, Safflower,
- **Fiber crop** -Cotton, Jute, Sunhemp, Mesta

- **Forage crop** – Oats, Napier/elephant grass, Berseem, Lucerne

Sugar crops – sugarcane, sugar beet

Cereal crops

1. **Rice**

Scientific name – *Oryza Sativa*

- **Aus/Autumn rice** – rice Varieties **grown during pre-monsoon period** is called as Aus rice,
- **Aman/ kharif rice** – called as **winter rice** due to harvesting in winter
- **Boro rice** – Rice **grown in submerged land** grown during **jan – feb**
- **Rice inflorescence**- is known as **Panicle**
- Rice is a **self-pollinated short-day plant**
- **Cardinal temperature** – 30-32^{0}C
- **Hulling percentage**- 70-75%
- **Pusa basmati 1 world first high yielding dwarf variety developed by IARI**

Figure 1 – Rice

- **Disease**- Bacterial leaf blight and Tungro virus
- **Zinc deficiency is the major problem in intensively cultivated and saline sodic soils which cause Khaira disease**. To treat this 100 kg /ha $ZnSO_4$ should be applied in saline soil.
- **Iron toxicity is major problem in acidic soil**, **Phalguna variety of rice have some tolerance to iron toxicity.**
- **Weed control – propanil @ 8 lt in 400 to 600 litter** water after 6 to 8 days of transplantation
- **Butachlor** – a pre-emergence weedicide
- **Critical stage for water**
    - **Booting stage**- most critical
    - **Tillering stage** – 0-20 days
    - **Primordia growth to flowering 40 – 60 days**, 5cm water is must.

- **Method of rice cultivation** – **Dapoong method**, Direct seeded plantation, transplantation
- **Rice is a tropical and sub-tropical plant** that requires a minimum temperature of **22°C and a rainfall of over 100 cm**.
- **Golden Rice is a genetically modified variety of rice that has been engineered to contain beta carotene, a precursor of vitamin A**.

2. Wheat

- **Emmer wheat** – grown in south states like Maharashtra, Tamil Nadu, Karnataka etc. and called as a Rava wheat
- ***Triticum aestivum* wheat is commonly grown in India** is also called as bread wheat.
- ***Triticum durum***- called as **Marconi wheat** used to make semya, spaghetti etc.
- ***T. Vulgare*- called as bread wheat** typically grown in rainfed condition in Indo Gangetic plains

Figure 2 – Wheat

- **Flowering of wheat is called as spike**
- **Shelling percentage** – 60 %
- **Dwarf variety- Lerma rojo, Sonora**
- **Varieties –** Kalyan Sona, Hira, Moti, Jawahar
- **Phalaris minor is the most common weed** which can be controlled by tribunal, Isoprotouron @ 2 kg per hectare 32-35 days after sowing
- **Late sown wheat varieties** – Sonara 64, Sarbati Sonora, Sonalika, Safed lerm.
- **Critical stage of irrigation- CRI stage (crown root initiation) 21 days after sowing**, **Flowering stage 90 DAS (day after set), Late tillering stage, Milk stage, Dough stage**
- **If water is available for 1 irrigation**, then **apply water at CRI stage**, if water is available for two irrigations, **then apply water at CRI stage and boot stage**, if three irrigations are available then **apply water at CRI, Boot stage, and milk stage.**

- Wheat is primarily grown in the temperate zones of the country. It requires a winter **temperature of 10°C to 15°C** and a summer temperature of **21°C to 26°C during** ripening for optimal growth.
- The crop **needs around 75 cm of water and a rainfall of 50 to 75 cm**. Excessive rainfall can be harmful to wheat cultivation as **standing water can destroy the roots of the plant.**

3. Maize

- ***Zea mays*** – scientific name
- Maize is a cross-pollinated crop. Its ***male flower (inflorescence) is called as a tassel, and female flower is called as silk.***
- Maize protein is called as zein
- **Maize is a warm-season crop** that thrives in temperatures between 18°C and 27°C during the day and 14°C to 15°C at night**. It cannot tolerate frost**, so planting needs to be timed accordingly.

Figure 3 – Maize

- It requires a **rainfall of 50-100 cm and a temperature ranging from 21°C to 27°C.**

4. Bajra

- Most **drought tolerant crop** among cereals and millets
- Sensitive to **water logging and acidic soil**
- **HB1 is the first hybrid variety of bajra.**
- Bajra is cultivated in **hot and dry climatic** conditions in the northwestern and western parts of India.

Figure 4 – Bajra

- It thrives in areas with a **rainfall of about 40-50 cm** and a **temperature range of 25°C to 30°C.** Bajra is a hardy crop that can withstand frequent dry spells and drought in these regions.
- It is grown both as a standalone crop and as part of mixed cropping systems**. Major producers of Bajra include Maharashtra, Gujarat, Uttar Pradesh, Rajasthan, and Haryana.**

5. Jowar (Sorghum)

- Jowar is the **main food crop** in the semi-arid areas of central and southern India. It is grown **as both a Kharif (summer) and a rabbi (winter) crop.**
- Jowar requires a mean monthly **average temperature of 26°C to 33°C during** the Kharif season and **around 30 cm of rainfall**. It can be cultivated in various soils, including loamy and sandy soils, but clayey, regur, and alluvial soils are most suitable.

Figure 5 – Jowar

- **Maharashtra is the leading producer of Jowar**, followed by Karnataka, Madhya Pradesh, and Andhra Pradesh. Jowar is also used as excellent poultry feed.
- **Roots are finer and more fibrous than maize**
- **Poor in lysine but rich in Leucine**
- Most important **male sterile variety is combine Kafir 60**

Pulse Crop

Pulses

Pulses are a **rich source of protein in India**, and the country is one of the leading producers of pulses globally.

Pulse cultivation is concentrated in the dryland areas of the Deccan, **central plateaus**, and northwestern regions of India.

Gram (chickpea) and tur (pigeon pea) are the main pulses cultivated in India. Gram is predominantly grown in subtropical areas during the rabi season in central, western, and northwestern parts of the country.

Tur is cultivated in marginal lands under rainfed conditions in the dry areas of central and southern states.

Major producers of gram are Madhya Pradesh, Uttar Pradesh, Maharashtra, Andhra Pradesh, and Rajasthan, while tur is primarily cultivated in Maharashtra, **Uttar Pradesh, Karnataka, Gujarat, and Madhya Pradesh.**

1. **Gram**

- Specific **sour taste of gram leaf is due to Maleic acid and oxalic acid**
- Gram seed should be sown at a **depth of 8 to 10 cm** in soil to escape from wilt disease.
- **Verities**- Gaurav, Avrodhi,
- **Sowing time – first fortnight of December**

Figure 8 – Gram

2. **Soyabean**

- **Called as wonder crop and meat for the poor**
- Due to the presence of **lipo-oxidase soyabean** is **not used as a Dal** as it produces off flavour.
- **Currently yellow seeded variety is more cultivated** than black seeded variety as it is not popular due to susceptibility to disease, low yield and highly shattered pods.

Figure 7 – Soyabean

Oilseeds

1. Groundnut

- **Contains 44 to 50 % oil content**
- **Peg is the gynophore** which bends downwards and forces the ovary into the soil. It is only after entering the soil **that ovary begins to develop into groundnut pods**.
- **Shelling percentage- 70%**
- Production **of Aflatoxin from Aspergillus flavus** is the major problem in groundnut post-harvest losses if stored under moist condition
- **Seed rate – 100 to 120 kg/ ha**
- **Plant population – 3.33 lakh per hectare**
- **Spacing should be** – 30 to 10 cm
- **Critical stage o irrigation**
  - **Flowering** – 35 to 40 days after set
  - **Pegging-** 55 days after set
  - **Pod formation** -65 – 70 days after set

Figure 8 – Groundnut

- It is a **tropical crop that requires 20°-30°C temperature and 50-75 cm rainfall**.
- It is mainly a **kharif** crop **but it also cultivated during rabi season**.
- It is **highly susceptible to frost**, **prolonged drought, continuous rain & stagnant water**.
- **Dry winter is needed at the time of ripening**.
- **Well drained** sandy loams, red, yellow and black cotton soils are well suited.
- **Rabi groundnut has a higher yield than that of kharif** due to better management practices and **less incidence of insect pest and disease**

2. Rapeseed And Mustard

- **Pungency is due to isothiocynate, enzymic hydrolysis product of glucosinolates** in all crucifers.
- **Duration of crop- 115-125 days**

- **Yield**- 1900-2500 kg/ha
- **Oil percentage**- 40%
- **Different species**- **Sarson, Toria (Rape), Rai (mustard), Taramira**, **white mustard**

Figure 9 – Mustard

Fiber Crop

1. Cotton

- **Botanical name**- ***Gossypium Hirsutum***
- ***Gossypium Hirsutum***- American cotton, it covers **approximately 50% area in India**
- **One bale – 170 kg of cotton**
- **Seed cotton**- Seed + lint
- **Cotton seed**- seed after removing lint
- **Topping is an important practice in cotton crop where removal** of terminal growing point once from each plant at 80 to 90 days after set to protect terminal growth and to encourage branching.

Figure 10 - Cotton

- **Hybrid 4 is the first commercial hybrid cotton of the world**
- **Bt cotton is the only genetically modified cotton variety** which is allowed to grow in India due to resistance against pink boll worm.
- **Cotton is a tropical crop primarily grown during the Kharif season in semi-arid areas of the country. India cultivates both short staple (Indian) cotton** and long staple (American) **cotton known as 'narma'** in the northwestern regions.
- **Cotton requires clear skies during the flowering stage** and thrives in an ideal temperature range of **21°C to 30°C, with a rainfall of 50 to 100 cm**.
- **Soil depth of 100-120 cm** is ideal for cotton cultivation.

- **Ginning percentage is 24-43%** in different cotton.
- **Nepiness – Sometimes fiber thickness is not uniform** and knots present on the fiber are also not distributed uniformly and it is referred as a low-grade cotton.
- **A good quality cotton fiber has all the knots properly distributed at equal distance and fiber is stronger.**

2. Jute

- **Scientific name** – ***Corchorus Capsularis*** – **hardy plant**, tolerate water logging. Its fiber is white in Colour hence called as white jute.
- **Planting time late feb to march** in low and mid lands.
- **Jute is a cash crop primarily cultivated in West Bengal** and adjoining eastern parts of the country.
- **West Bengal contributes to about three-fourths of the jute production in India**, while Bihar and Assam are other significant jute-growing regions.
- Jute grows **best in temperatures ranging from 24°C to 35°C and requires a rainfall of 120 to 150 cm.**

Figure 11 – Jute

- **Ideal state for harvesting -small pod stage or initiation of pod formation that is 135 to 140 DAS (days after set.**
- **Steeping (Soaking) – After 2-4 days of harvesting** the plants are shaken for complete leaf shedding and then tied into bundles. **After 3 to 4 days of tieing in bundle should be submerged in water for retting.**
- **Retting- it is a microbial degradation by which the microbes degrade the cellulosic material around bast fiber** and facilitate separation of fiber from the stem. **Retting needs 34^{0}C Water temperature** and completed **with in 10-15 days during July, 18 to 20 days during aug and September and 2**1-30 days after September.
- **After retting the fiber is extracted by beating**.

Sugarcane

Scientific name

a- **Tropical cane** -***Saccharum Officinarum*** – **Thick and juicy canes, good for chewing purpose indigenous to New guinea, high sugar content**, low fiber and restricted to tropical areas.

b- **Indian cane** – ***Saccharum Barberi*** – Indigenous to north eastern India, short and thin stalks, early maturity, low to medium sucrose content

c- ***Saccharum Sinense***- **Indigenous to north eastern India**, long and thin stalks, low to medium sucrose content and early maturity.

Figure 12 – Sugarcane

- **Inflorescence of sugarcane is open panicle and generally called as arrow.**
- Upper **1/3rd part of the cane is used for planting purpose**.
- After production 50% of sugarcane can be utilized for white sugar, 30% low purity sugar for jaggery production and 20% for alcohol production.
- **Sugarcane breeding institute in Coimbatore**
- Indian institute of sugarcane research in Lucknow
- **Growth condition for high tillering**
  - **High Heat requirement 25-30^{0}C**
  - High **moisture requirement – 77 – 88 %** of field moisture capacity.
  - **Good illumination**- Bright sunshine and water availability.
- **Gasohol is prepared from 80% petrol + 20% alcohol (from** sugarcane) which is used in automobiles.
- **Higher dose of Nitrogen decreases sucrose content.**
- **Method of planting** – Flat bed planting, Trench/Java method, Partha method, Spaced transplanting, winter nursery system, Sablang/ sprouting method.
- **Chemical ripener like Polaris improve sucrose** when sprayed on foliage 6 weeks before the scheduled harvest.
- **Duration of crop** – 10 to 12 months
- **Sucrose % 14 – 17 %**

- **Sugarcane is unsuitable to grow in saline soil**
- We can get 1 crop in a year.

Tea

- **Tea is a plantation crop primarily cultivated for use as a bever**age. It is grown in hilly areas with undulating topography and well-drained soils in the humid and sub-humid tropics and sub-tropics.

Figure 13 – Tea

- **The ideal temperature range for tea growth is between 20°C and 30°C, and it requires a rainfall of approximately 150-300 cm**.
- In India, tea plantations **are mainly found in the Brahmaputra valley of** Assam, the sub-**Himalayan region of West Bengal** (Darjeeling, Jalpaiguri, and Cooch Behar districts), and the lower slopes of the Nilgiris and Cardamom hills in the Western Ghats.

Coffee

- ***Coffee is a tropical plantation*** crop that is cultivated in the highlands. There are three main varieties of **coffee: arabica, robusta, and liberica**.

Figure 14 – Coffee

- In India, **coffee is primarily grown in the highlands of the Western Ghats, spanning Karnataka, Kerala, and Tamil Nadu**.
- The **ideal temperature range for coffee growth is between 15°C and 28°C,** and it **requires rainfall ranging from 150 cm to 250 cm**.

Tobacco

Two Indian tobacco spp

- ***Nicotiana Tabacum***- **Plants height 150-250 cm**, large and narrow leaf, texture of leaf is fine**, nicotine content is 0.5-5.5%,** used for smoking and chewing purpose, grown on light and high lands, seeds rate is higher than rustica.
- ***Nicotiana Rustica***- **Plant height 90-120 cm nicotine content is 3.5 – 8.0%** used for hookah, chewing and snuff purpose, grown in heavy, low-lying soils.

Figure 15 - Tobacco

- **Sodic soil is unfit for tobacco production** because the plants absorb a lot of chloride ions which result in poor burning quality of leaves and a mild acidic soil.
- **Nicotine is produced mainly in plants roots** and is carried through stems to leaves where it is stored.
- Nursery is prepared 2nd fortnight of august.
- **Transplanting age – 4 to 5 leaves**
- **Topping** – **Removal of flower heads either alone or with few upper** or top leaves from plants to improve the size, body and quality of leaves. It improves the branching.
- **De-suckering**- After topping, **axillary buds grow so the removal of** such lateral branches or suckers or auxiliary buds is called as De-suckering.
- **The main aim of topping and de-suckering is to divert energy and nutrients from flower head to leaves.**
- **Priming**- the removal of lower leaves which comes in contact with soil and lose their commercial value.
- **Curing-** It is the **process of drying most of the moisture of leaf's is removed to impart required Colour, texture, and aroma to the final product**.